

Nama : Calvin Wiliam Wijaya

Kelas : TRPL 1B

Mata Kuliah : PBO 1

Jawaban Soal IP Address (No 1–7)

Soal 1: 10.51.74.65/12

1. Byte pertama: 10
2. Jenis Class: A
3. Private/Public: Private
4. Prefix: /12 (12 bit network)
5. Host bits: $20 \rightarrow 2^{20} - 2 = 1.048.574$ host usable
6. Subnet Mask: 255.240.0.0
7. Oktet dipengaruhi: 2
8. Block size: 16
9. Network: 10.48.0.0
10. Broadcast: 10.63.255.255
11. Host range: 10.48.0.1 – 10.63.255.254

Soal 2: 192.168.30.145/28

1. Byte pertama: 192
2. Jenis Class: C
3. Private/Public: Private
4. Prefix: /28 (28 bit network)
5. Host bits: $4 \rightarrow 2^4 - 2 = 14$ host usable
6. Subnet Mask: 255.255.255.240
7. Oktet dipengaruhi: 4
8. Block size: 16

9. Network: 192.168.30.144

10. Broadcast: 192.168.30.159

11. Host range: 192.168.30.145 – 192.168.30.158

Soal 3: 192.168.215.122/27

1. Byte pertama: 192

2. Jenis Class: C

3. Private/Public: Private

4. Prefix: /27 (27 bit network)

5. Host bits: $5 \rightarrow 2^5 - 2 = 30$ host usable

6. Subnet Mask: 255.255.255.224

7. Oktet dipengaruhi: 4

8. Block size: 32

9. Network: 192.168.215.96

10. Broadcast: 192.168.215.127

11. Host range: 192.168.215.97 – 192.168.215.126

Soal 4: 172.31.22.200/16

1. Byte pertama: 172

2. Jenis Class: B

3. Private/Public: Private

4. Prefix: /16 (16 bit network)

5. Host bits: $16 \rightarrow 2^{16} - 2 = 65.534$ host usable

6. Subnet Mask: 255.255.0.0

7. Oktet dipengaruhi: 3

8. Block size: 256

9. Network: 172.31.0.0

10. Broadcast: 172.31.255.255

11. Host range: 172.31.0.1 – 172.31.255.254

Soal 5: 192.168.200.215/26

1. Byte pertama: 192
2. Jenis Class: C
3. Private/Public: Private
4. Prefix: /26 (26 bit network)
5. Host bits: $6 \rightarrow 2^6 - 2 = 62$ host usable
6. Subnet Mask: 255.255.255.192
7. Oktet dipengaruhi: 4
8. Block size: 64
9. Network: 192.168.200.192
10. Broadcast: 192.168.200.255
11. Host range: 192.168.200.193 – 192.168.200.254

Soal 6: 172.20.20.20/14

1. Byte pertama: 172
2. Jenis Class: B
3. Private/Public: Private
4. Prefix: /14 (14 bit network)
5. Host bits: $18 \rightarrow 2^{18} - 2 = 262.142$ host usable
6. Subnet Mask: 255.252.0.0
7. Oktet dipengaruhi: 3
8. Block size: 4
9. Network: 172.20.0.0
10. Broadcast: 172.23.255.255
11. Host range: 172.20.0.1 – 172.23.255.254

Soal 7: 10.168.201.200/13

1. Byte pertama: 10
2. Jenis Class: A

3. Private/Public: Private
4. Prefix: /13 (13 bit network)
5. Host bits: $19 \rightarrow 2^{19} - 2 = 524.286$ host usable
6. Subnet Mask: 255.248.0.0
7. Oktet dipengaruhi: 2
8. Block size: 8
9. Network: 10.168.0.0
10. Broadcast: 10.175.255.255
11. Host range: 10.168.0.1 – 10.175.255.254